(Please write your Enrollment Number) Enrollment No.

MID-TERM EXAMINATION
(B.Tech. CSE-AI/AI &ML)
1st semester(January, 2023)
OFFLINE mode

Subject Code: BAI-110	Subject: Programming With Python

Time : 1 ½Hours Maximum Marks : 30

Q1 Attempt all Parts **(2.5*4)**

(a) Write a Python program to count the number of even and odd numbers from a series of numbers.
(b) Explain different types of datatypes in python.
(c) Write a program to check if the word 'open' is present in the "This is open source software".
(d) Explain with examples different types of list operators in python.

Q2 (Attempt any Two Parts) **(5,5)**

(a) Explain various types of operators in python. Write a program to find the square of a number.
(b) With the necessary examples describe the use of continue and break statement. Write a Python program that prints all the numbers from 0 to 6 except 3 and 6.
(C)Write a python program to print the following alphabet pattern :

```
ABCDEFEDCBA
 ABCDEDCBA
  ABCDCBA
   ABCBA
    ABA
     A
```

Q3 (Attempt any Two Parts) **(5,5)**

(a) What is a local variable? How is it different from global variable? Write a python function to find the factorial of a non-negative number.
(b) Write a Python program to print the even numbers from a given list.
Sample List: [1, 2, 3, 4, 5, 6, 7, 8, 9]
Expected Result: [2, 4, 6, 8]
(c)Write the python program the print the given series : 14,28,20,40,32,64,50,100